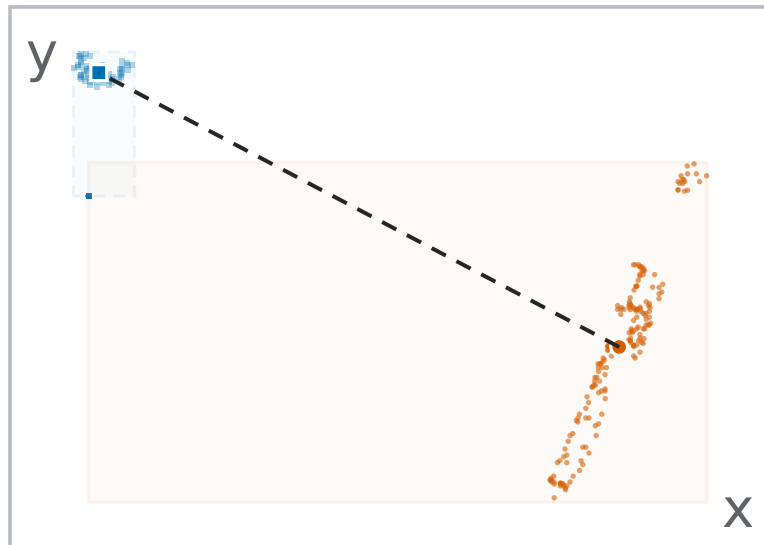


(a) Failure example



heater close by trash can

Open3DSG: $Z=.853$; rank 19

XY distance: 4.33 m

(b) Compatibility and score

Predicate T
close by

Geometry G
distance / dz
overlap

Score Z
ranking only

Compatibility
 $C^{tr}(T,G)$

linked-pair ordering
transformation averaging

$C^{tr}(T,G)=0.003$

Within-family score
 $u = Z \times C^{tr}(T,G)$

(c) Re-ranking

19

source rank

178

RelCompat3D

**Family-aware
re-ranking**

proximity / vertical
 $Z \times C^{tr}$
support/contact
keep source order

Joint evaluation

Recall@K $\uparrow$ Violation@K $\downarrow$